Architecture Agent: diagram pipeline revamp

Before and after. Goal: the most accurate diagram, at the lowest time and cost, with no AWS services the ADR did not decide on.
October 17, 2026

0	4.9 s	~10 s	0	0
model calls per diagram (was 3–6 Sonnet calls)	17-box diagram, end to end (was 2–7 min)	36-box, two-AZ diagram 37/37 edges routed	layout errors, 0 retries (was up to 2 LLM retries)	services added that are not in the ADR design

Pipeline

Before

`adr.md → LLM Extractor (+repair) → LLM Assumer → LLM Diagram coder (pixel coordinates, up to 2 retries) → XML → libavoid → export`

After (default)

`design.json → spec in code → ELK computed layout → XML → libavoid → export`

Document input (no design) still uses the Extractor and Assumer, with the new limits on services.

Before and after

Area	Before	After
Diagram content	An LLM re-read the rendered `adr.md` prose, so it could drop, rename or invent components. Nothing compared the diagram with the ADR design.	The ADR's own components, groups and flows are mapped in code (`design_spec.py`). The diagram matches the ADR exactly.
Unwanted AWS services	Blocked by prompt rules only. The Assumer's additions were drawn, and inferred services could slip in.	From a design: none can be added. From documents: inferred AWS services the documents never name are dropped, and the Assumer's additions are reported as `suggestions` but not drawn.
Layout	The LLM guessed absolute pixel positions. Overlaps and containment errors triggered retries, and leftovers shipped as warnings.	ELK (elkjs 0.12.0, pinned) computes positions and nested group sizes. Boxes cannot overlap or leave their group, and the same input always gives the same layout.
Evidence checks	An assumption with a quote not found in the documents was kept, with a warning.	That assumption is dropped.
Icons	Unmatched services silently got a generic icon (`matched=False` was never read).	Each generic-icon fallback is a warning in the report.
Geometry after fixers	The vendored fixers could move boxes after the layout check, unchecked.	Overlaps are checked again on the finished drawing.
Large PNG export	draw.io silently cut off big PNGs: the 36-box diagram was blank below 76% of its height.	PNG scale drops automatically to stay under about 20 Mpx; the 36-box export renders in full.
API	`generate_diagram` accepted `adr_md` only.	Accepts `design` (preferred) or `adr_md`. `run_pipeline` passes `design.json` automatically.
Report	Status, steps, warnings.	Adds `input`, `layout_engine`, `model_calls` and `suggestions`.

Settings (old behaviour still available)

Setting	Default	Options
`LAYOUT_ENGINE`	elk	llm (LLM coder; also the automatic fallback)
`ASSUMPTIONS_MODE`	suggest	draw, off (skips one model call)
`NAMED_SERVICES_ONLY`	true	false

Verification

- All 340 existing tests pass; 3 were updated for intended changes.
- Real agent action run from a design with invalid AWS credentials: status `pass`, 0 model calls, 0 tokens.
- 17- and 36-box diagrams rendered and reviewed: 0 layout errors, every edge routed.
- Fake-LLM document run: an injected, never-named AWS WAF was dropped; the Assumer's API Gateway was listed as a suggestion, not drawn.

Open items

- Not verified: the ARM64 Docker build and real Bedrock calls on the document path. `SYSTEM_DESIGN.md` still describes the old flow. No new tests were added for the new paths.
- Known limits: dense diagrams can still have some crossing lines, and a flow that runs backwards in the design is drawn leftward.
- UI action: send `design` to `generate_diagram` to get the zero-call path.

Files: `diagram_generator/design_spec.py`, `render/elk_layout.py`, `render/layout_elk.mjs`, `stages.py`, `harness/h2_assumptions.py`, `harness/layout_check.py`, `render/drawio_tools.py`, `catalog/loader.py`, `config.py`, `architecture_agent/actions.py`, `vendor/elkjs/`. Pre-change backup: `.revamp_backup.tar.gz`.